Arduino && WebSockets

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Check-In

Which Homer are you?



Goal

Use an Arduino to communicate instantly with a Node server using web sockets.

Agenda

- Background 10 min
- Setup 5 min
- Arduino to Arduino 20 min
- Browser to Arduino 20 min
- Questions 5 min

Today

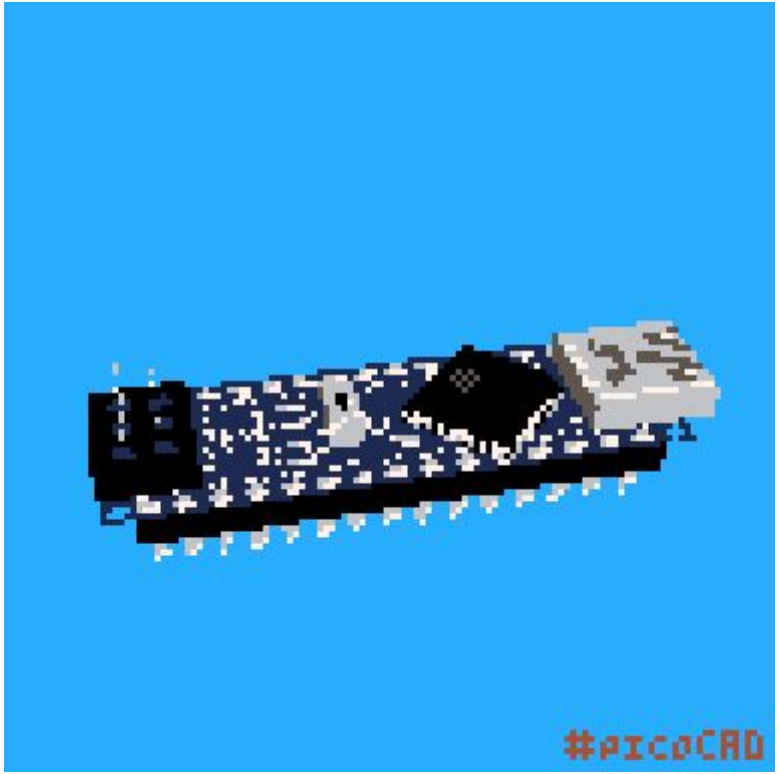
Two simple examples!

1. Arduinos instantly talking to each other through web server, no frontend.
2. Web frontend instantly controlling distributed Arduinos.

Why?

Lets a website control an arduino

Lets an Arduino control a webpage



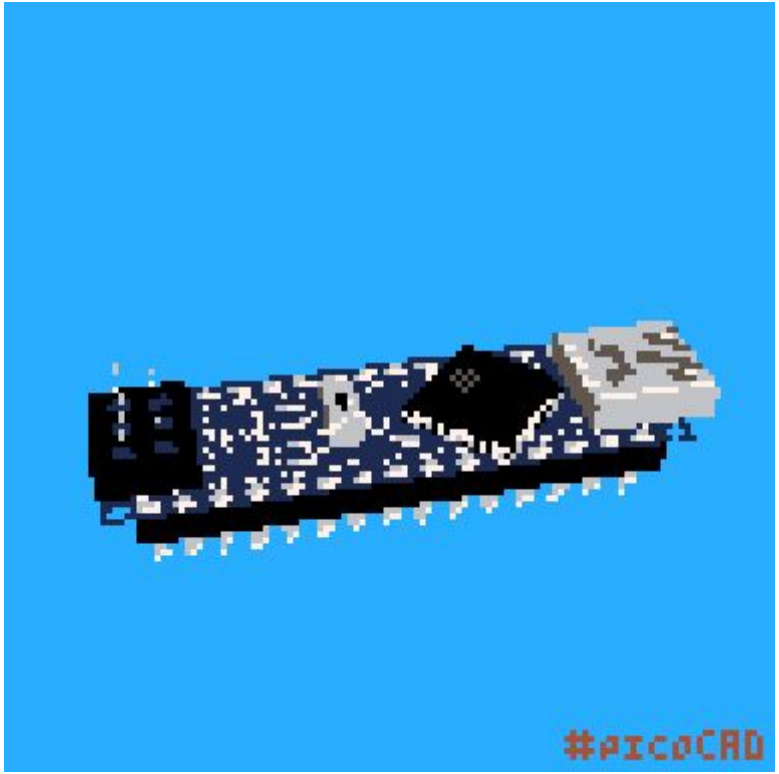
Why?

Lets a website control an arduino

- Turn a motor
- Switch lights on/off

Lets an Arduino control a webpage

- Send live sensor data to a website
- Different interfaces for website control.

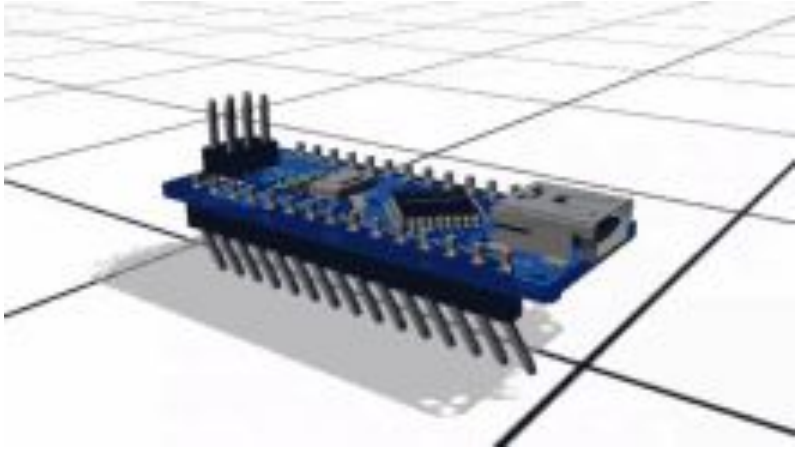


Why WebSockets?

A phone call, not a letter.

Regular web requests are like sending letters: you ask, wait, and get one reply. **This is called polling.**

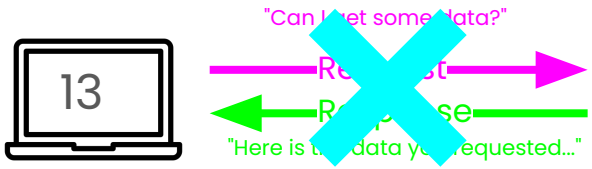
A WebSocket is an open phone line: both sides can talk anytime, instantly. **No polling delays.**



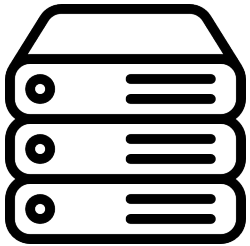
HTTP Server Flow



HTTP



Server

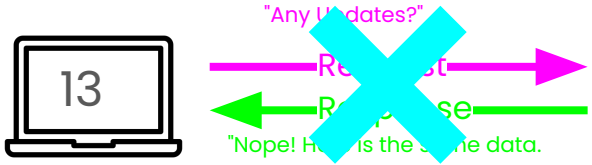




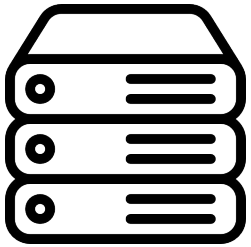
```
data: {[
  "Sensor":13
]}
```



HTTP



Server



data: {[
"Sensor":13
]}

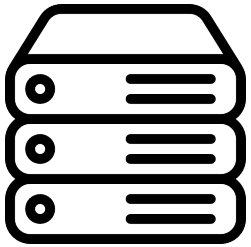


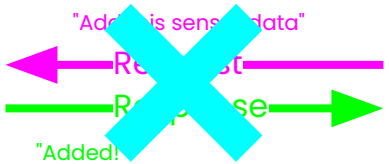


HTTP



Server







```
data: {[
   "Sensor":13
   "Sensor":505
]}
```

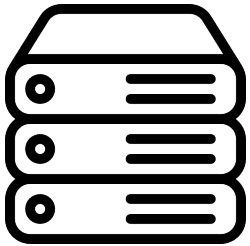


HTTP

505

"Yeah here is the data you requested!"

Server



data: {[
"Sensor":13
"Sensor":505
]}



WebSockets Flow



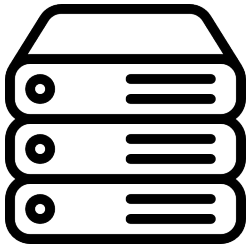


WebSockets

Server









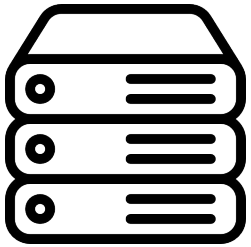


WebSockets

Server



Socket



Socket



"New client connected!"

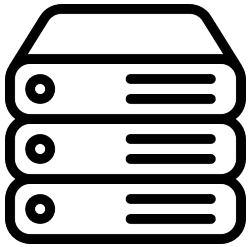


WebSockets

Server



Socket



Socket



"New sensor data!"

"Broadcast sensor data!"



WebSockets



"Light on!"

Socket

Server

Socket



"Button Pressed!"



"Light on!"

Socket

Socket



"Light on!"



"Light on!"

Socket

Socket



"Light on!"



WebSockets



"New Brightness!"

Socket

Server

Socket



"New Brightness!"



"Change Brightness!"

Socket

Socket



"New Brightness!"



"New Brightness!"

Socket

Socket



"New Brightness!"

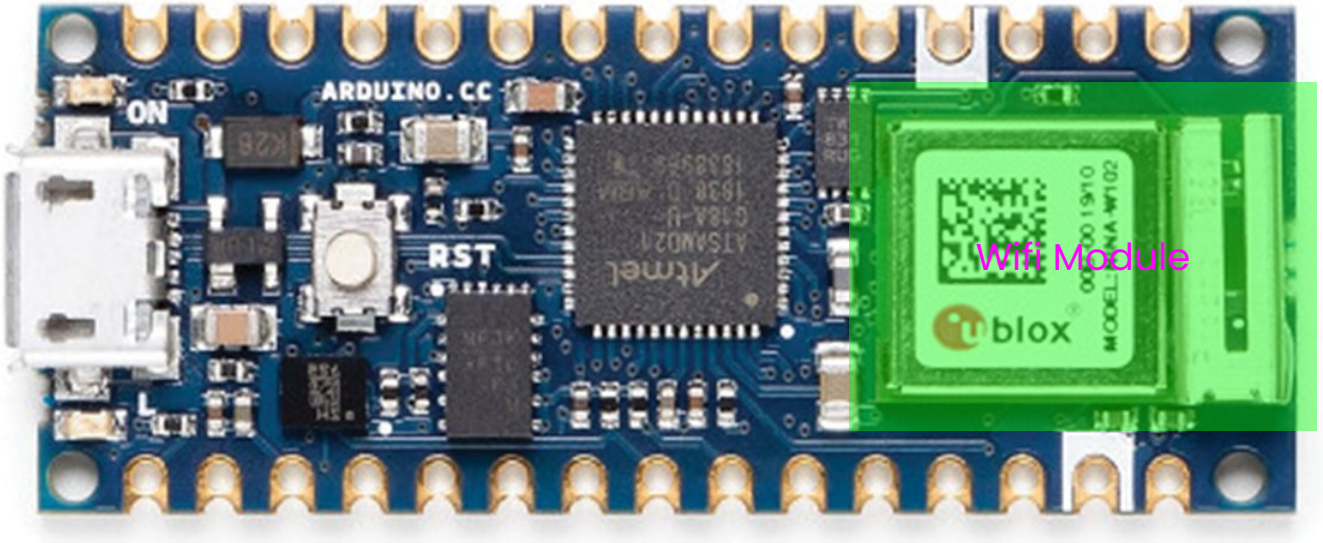
So we need...

- Arduino connected to the internet
- Server with routes + sockets
- Arduino connection to socket server
- Arduino to send messages to server
- Server to handle sockets messages
- Arduino to reacts to new messages

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Arduino Nano IOT



Import WifiNINA Library

Add Library in Arduino IDE

Connect to Wifi

`arduino_code/ex0_wifi-tester`

So we need...

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Server

Read through `server.js` file.

What are some things that look familiar?

5 min

Why no socket.io?

Why no socket.io?

socket.io is Node + frontend library for socket library.

It provides some sugar for making sockets easier and has cool features (rooms, polling fallback, etc)

BUT it doesn't work with Arduino without some extra work that isn't worth it.

Instead we will use a plain ol' web socket connection with a different Node library.

So we need...

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Import WebSocketClient

Add Library to the Arduino IDE

Socket Handshake

`arduino_code/ex1_turn-on-the-lights`

line 145: `connectWebSocket();`

So we need...

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Arduino Sends Message

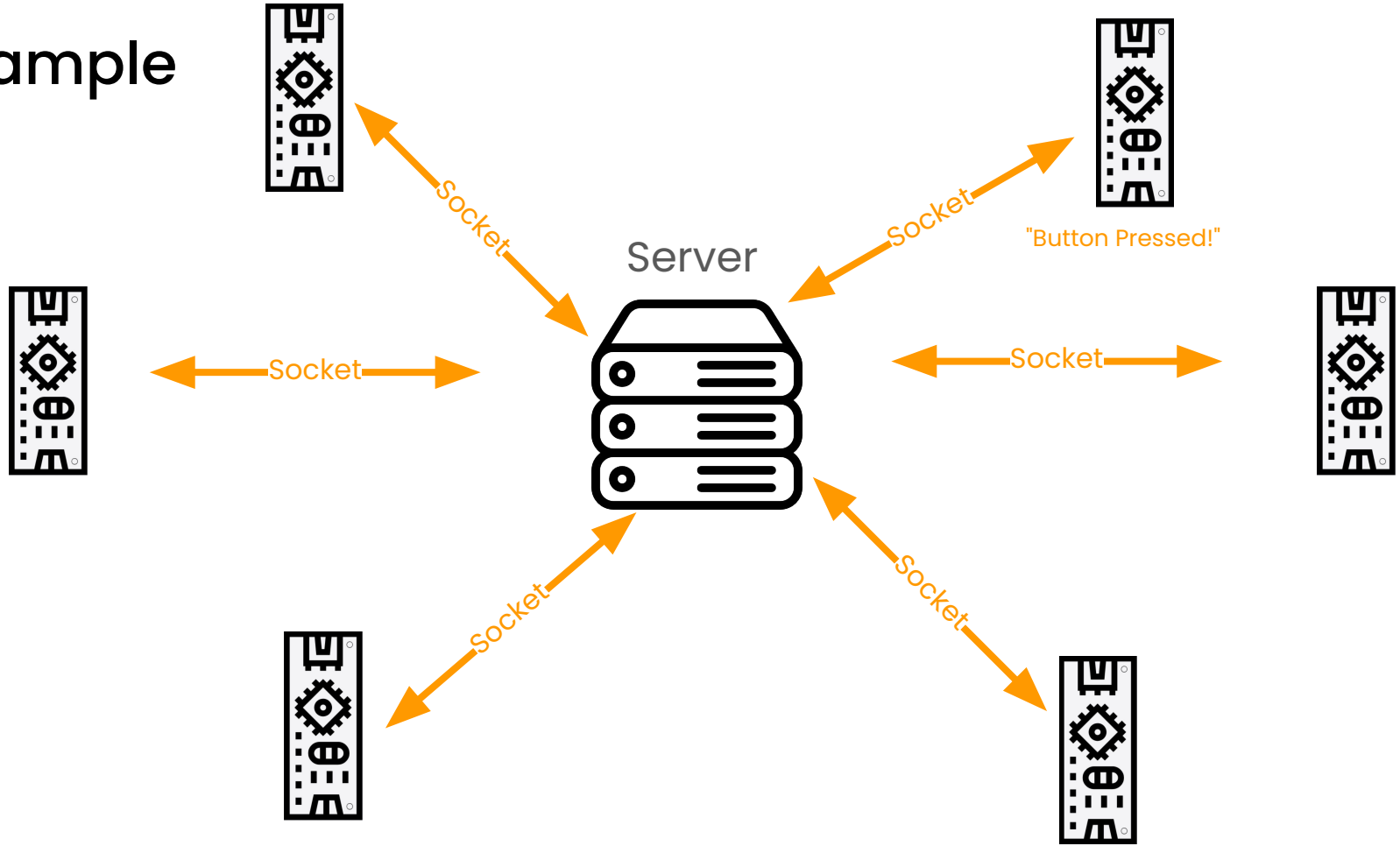
`arduino_code/ex1_turn-on-the-lights`

line 89: `handleMessage();`

So we need...

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- ~~Arduino connection to socket server~~
- ~~Arduino to send messages to server~~
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Example 1


Server
Socket
Socket
Socket
"Button Pressed!"
Socket
Socket
Socket

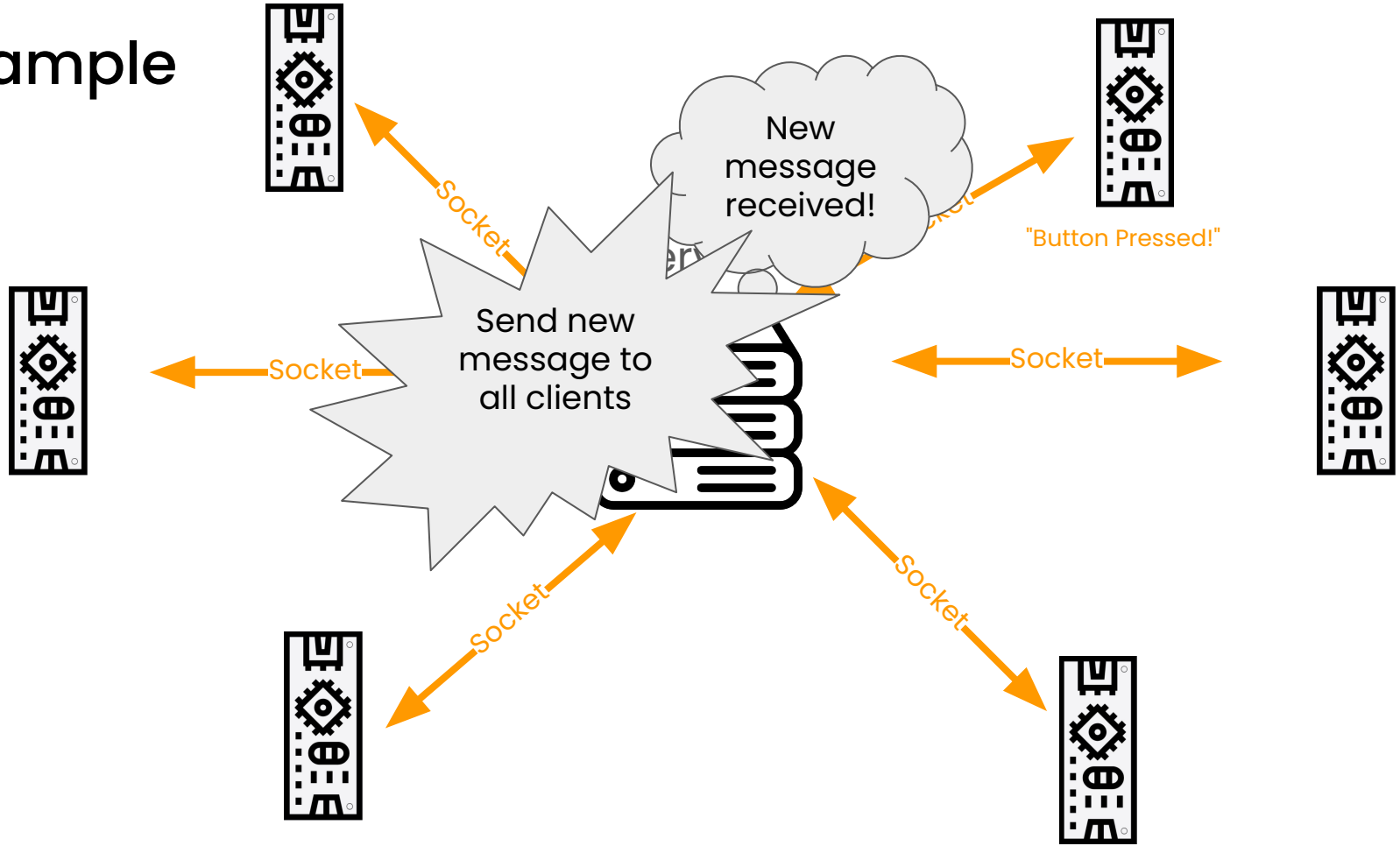
Server

Look again at `server.js` file.

On what line does the server handle the code?

5 min

Example 1


Socket
New message received!
"Button Pressed!"
Socket
Send new message to all clients
Socket
Socket
Socket

So we need...

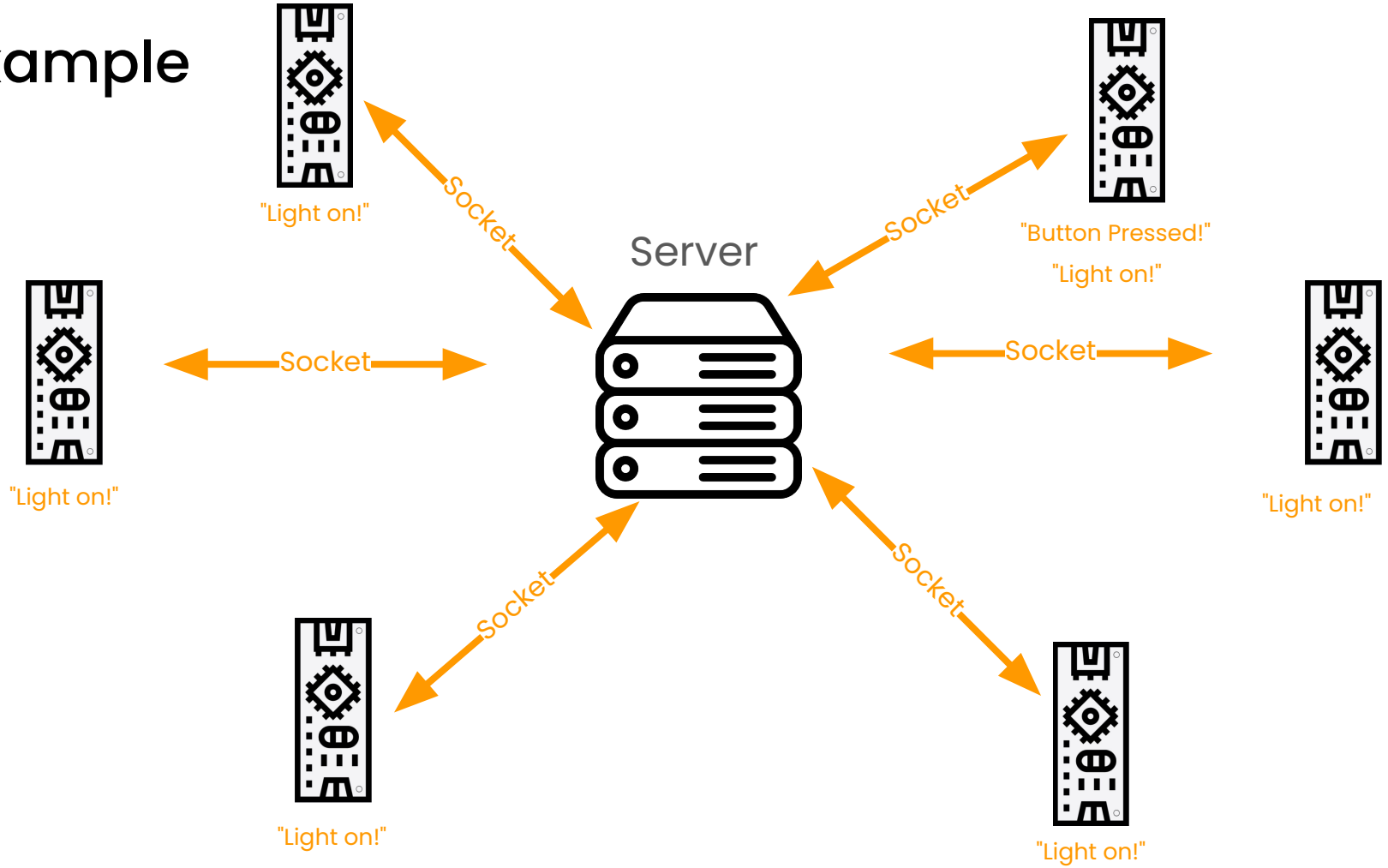
- ~~Arduino connected to the internet~~
- ~~Server with routes + sockets~~
- ~~Arduino connection to socket server~~
- ~~Arduino to send messages to server~~
- ~~Server to handle sockets messages~~
- Arduino to reacts to new messages

Arduino Reacts

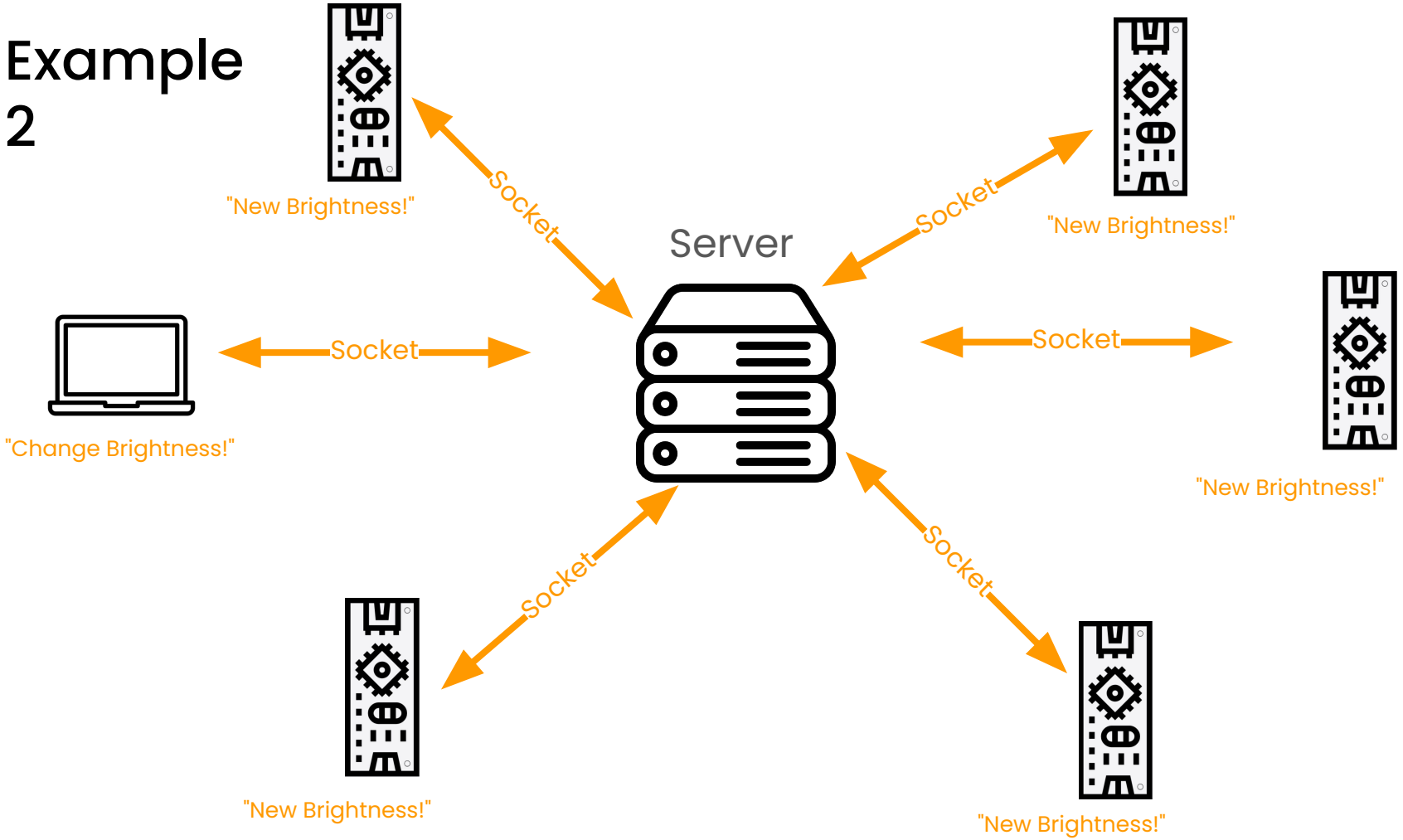
`arduino_code/ex1_turn-on-the-lights`

line 163: `handleMessage();`

Example 1


"Light on!"
Socket
Server
Socket
"Button Pressed!"
"Light on!"
"Light on!"
Socket
Socket
"Light on!"
Socket
Socket
"Light on!"
"Light on!"

Example 2


"New Brightness!"
Socket
Server
Socket
"New Brightness!"
"Change Brightness!"
Socket
Socket
"New Brightness!"
Socket
Socket
"New Brightness!"
"New Brightness!"

More than one type

`arduino_code/ex2_turn-on-the-lights`

line 188: `handleMessage();`

Challenge

Combine the two sketches!

Connect a force sensor to the Arduino

Write a p5 sketch the uses the force the change the size of the circle.

The higher the force the smaller the circle.

Hint - You'll either have to change and deploy your own server or highjack one of my value, like brightness

More Ways to Connect

Arduino Connection Options

Method	Range	Power	Built in?
WiFi / HTTP	Short to medium	High	✅
BLE	Short	Very low	✅
LoRa	Long	Low	🚫
ESP-NOW	Medium	Low	🚫 esp32 only
Cellular	Short to very long	Very high	🚫

Questions?